

Synopsis

Problem Statement

Flight Booking System (Customer Module)

Project ID: 24

Student Name:

University Roll number:

Roll No:

Section: A

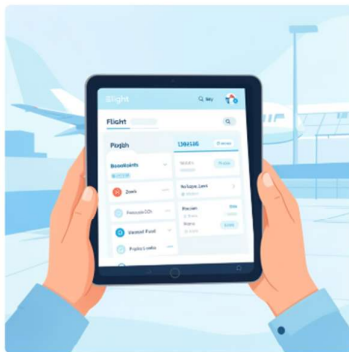
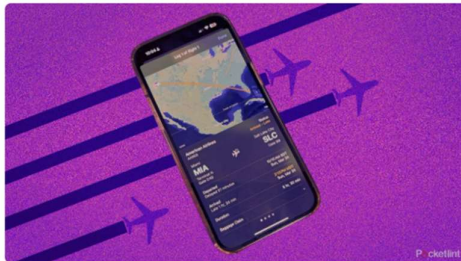
Semester: 2

Problem Description:

- The Flight Booking System (Customer Module) is designed to make booking flights simple, fast, and secure for users.
- It allows customers to search for flights by destination or flight number and sort them by departure time or ticket price, helping them choose the best option according to their needs and budget.
- Once a suitable flight is selected, users can enter passenger details and payment information to confirm their booking.
- The system also includes user authentication to ensure that personal and booking information remains safe.
- If plans change, customers can easily cancel their reservations. All booking data is securely stored and updated using file handling, ensuring accurate records.

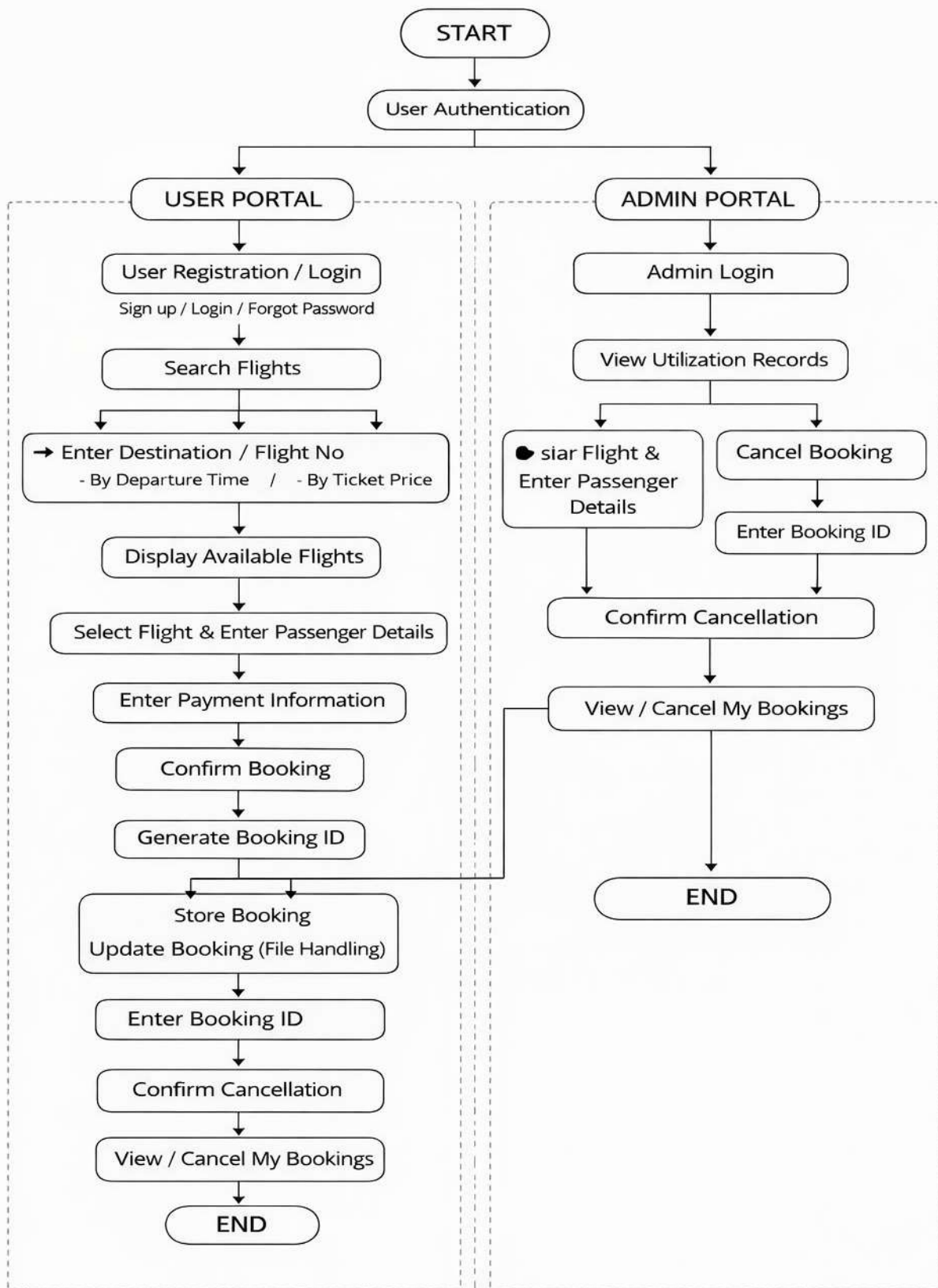
Role in Real Life:-

- In real life, such a system saves time, reduces manual paperwork, and makes travel planning more convenient. It allows people to book tickets anytime from anywhere, compare prices quickly, and manage their trips efficiently.
- This system plays an important role in modern digital travel services. It also improves customer satisfaction by providing instant booking confirmation, transparent fare details, and quick access to travel history, making the entire process more reliable and user-friendly.



Steps of Implementation:

Flight Booking System – Customer Module



1. Requirement Analysis

- Identify the main functionalities: user registration and login, flight search, sorting, booking confirmation, booking cancellation, and secure data storage using file handling in C.

2. System Design

- Divide the system into modules such as authentication, flight search, booking, cancellation, and file management.
- Design structured records for users, flights, and bookings to organize data properly.

3. User Authentication

- Implement registration to store user credentials in a file.
- Implement login to verify credentials by reading stored data.
- Allow access to the system only after successful authentication.

4. Flight Search and Display

- Load flight records from a file.
- Allow users to search by destination or flight number.
- Provide sorting options based on ticket price or departure time.
- Display available flights that match the search criteria.

5. Booking Process

- Allow the user to select a flight.
- Collect passenger details and payment information.
- Generate a unique booking ID.
- Store booking details in a file and update seat availability.

6. Booking Cancellation

- Allow users to enter their booking ID.
- Verify the booking from stored records.
- Remove or mark the booking as cancelled.
- Update seat availability accordingly.

7. File Handling

- Use files to store and manage user, flight, and booking data.
- Ensure proper opening, reading, writing, and closing of files to maintain data accuracy.

8. Testing and Integration

- Test each module individually.
- Integrate all modules into a main menu-driven program.
- Verify proper navigation, data consistency, and error handling.

Proposed Modules:

1. User Authentication Module

- Handles user registration, login, and logout.
- Stores and verifies user credentials using file handling.
- Uses C concepts:
 - struct to store user details
 - FILE pointers (fopen, fread, fwrite, fprintf, fscanf)
 - String handling functions (strcmp)
 - Conditional statements (if-else) for validation

2. Flight Search Module

- Reads flight data from flights.txt.
- Allows searching by destination, date, or flight number.
- Provides sorting by price or departure time.
- Uses C concepts:
 - struct for flight records
 - File reading functions (fopen, fread, fscanf)
 - Arrays to store multiple flight records
 - Loops (for, while) for searching
 - Sorting algorithms (Bubble Sort / Selection Sort)
 - String comparison functions

3. Flight Booking Module

- Collects passenger details.
- Allocates seats and confirms booking.
- Generates a unique booking ID.
- Updates booking records in bookings.txt.
- Uses C concepts:
 - struct for booking details
 - File writing (fwrite, fprintf)
 - Random number generation or counter for Booking ID

- Loops and conditional statements
- Updating seat availability in file

4. Payment & Billing Module

- Accepts payment details.
- Validates payment input.
- Generates receipt after successful transaction.
- Uses C concepts:
 - Input/output functions (scanf, printf)
 - Conditional validation (if-else)
 - File writing for receipt generation
 - Basic arithmetic operations for total fare calculation

5. Booking Management Module

- Allows users to view, update, or cancel bookings.
- Searches booking file using Booking ID.
- Updates or removes records.
- Uses C concepts:
 - File handling (read, write, append modes)
 - Temporary file technique for record deletion
 - Loops for searching records
 - Conditional statements for verification

6. File Handling / Data Storage Module

- Manages storage of users, flights, and bookings.
- Ensures data consistency and accuracy.
- Uses C concepts:
 - FILE structure
 - File modes (r, w, a, rb, wb)
 - Error handling (NULL file pointer checking)
 - Proper file closing using fclose()



Flight Booking System

User Authentication Module

Use C functions to handle user sign-up and login securely by reading and writing user data to 'users.txt' file



Flight Search Module



C reads flight data from 'flights.txt' to conduct searches by destination, date, or flight number and sort them by price or departure time



Flight Booking Module



C is used to take passenger details, allocate seats, and confirm booking by updating booking data in 'bookings.txt' file



Payment & Billing Module



Secure payment processing is handled in C, generating receipts in 'receipts.txt' file after successful payment.



Flight Booking Module



C is used to take passenger details, allocate seats, and confirm booking by updating booking data in 'bookings.txt' file



Payment & Billing Module



Secure payment processing is handled in C, generating receipts in 'receipts.txt' file after successful payment.



Booking Management Module



Utilize C to view, update, or cancel bookings efficiently by modifying 'bookings.txt' file

File Handling / Data Storage



Use C file handling to store and update booking records in 'users.txt' & 'bookings.txt' for reliable data storage and retrieval

Required topics from the subject:



Strings

Strings are used to store text such as passenger names, destinations, and flight numbers. Functions like `strlen()`, `strcmp()`, and `strcpy()` help in comparing and managing text data.



Structures (struct)

Structures are used to group related data like passenger name, age, flight number, and ticket price into one unit. This makes the program organized and easy to manage.



Functions

Functions divide the program into smaller reusable blocks like `login()`, `searchFlight()`, and `bookTicket()`. This improves readability and reduces code repetition.



Pointers

Pointers store the address of variables and help in dynamic memory management. They are useful for handling arrays, structures, and passing data efficiently to functions.



File Handling

File handling is used to store and retrieve booking records permanently. Functions like `fopen()`, `fwrite()`, `fread()`, and `fclose()` help manage customer data securely.



Conditional Statements & Loops

if-else, switch, for, and while loops help in decision-making and repeating tasks such as menu display and booking operations.

Platform Required:

The project can be developed using:

1. Visual Studio
2. Code::Blocks
3. Turbo C
4. Any C/C++ compatible compiler

Books and Link Sources:

1. Programming in ANSI C by E.Balagurusamy
2. Let Us C by Yashavant Kanetkar
3. The C Programming Language by Brian W. Kernighan & Dennis Ritchie
4. C Programming Absolute Beginner's Guide by Greg Perry & Dean Miller

Online Sources:

1. <https://www.geeksforgeeks.org>
2. <https://www.tutorialspoint.com>
3. <https://www.w3schools.com>
4. <https://www.javatpoint.com>